Karan Palariya

AI/ML & Software Engineering Intern Candidate — Bhimtal, Uttarakhand, India

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PROFILE

CS undergraduate (2027) building production-grade AI systems — retrieval-augmented generation, parameter-efficient fine-tuning, and full-stack ML platforms. Seeking Software Engineering / AI-ML / Backend internships where I can ship systems, not just notebooks.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL, HTML/CSS
AI/ML: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, RAG, embeddings, cosine-similarity retrieval, LoRA/PEFT fine-tuning, CNNs, NLP fundamentals, Computer Vision (OpenCV, MediaPipe)
Backend/Full-Stack: Flask, React.js, REST API design, MySQL
Cloud & Tools: AWS (Certified Cloud Practitioner), Oracle Cloud Infrastructure (Certified AI Foundations Associate), Git, GitHub, Jupyter
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Computer Architecture, Operating Systems, Compiler Design

PROJECTS

SimpleRAG — Retrieval-Augmented Generation Pipeline **2026**
PyTorch · Cosine Similarity · Embedding Retrieval

- Engineered a retrieval-augmented generation pipeline in PyTorch implementing L2-normalized cosine-similarity search and query-document embedding fusion, retrieving top-k passages with 85% top-k accuracy on a 500-document corpus.
- Reduced irrelevant-context retrieval by fusing query and document representations before ranking, cutting average query latency to ∼140 ms.
- Designed the module as a reusable retrieval component, later extended into the architecture for DocMind, an agentic enterprise RAG copilot.

LoRALinear — Parameter-Efficient Fine-Tuning Layer **2026**
PyTorch · LoRA · From-Scratch Implementation

- Implemented a LoRA linear layer from first principles using Kaiming-uniform initialization and a custom `nn.functional.linear` forward/backward pass.
- Cut trainable parameters by ∼92% versus full fine-tuning while retaining ∼96% of baseline task accuracy, demonstrating PEFT fundamentals.
- Verified correctness against a reference implementation with unit tests covering rank, alpha scaling, and gradient flow.

MediLineage — Full-Stack Healthcare Risk Prediction Platform **Aug 2025**
React.js · Flask · MySQL · Scikit-learn

- Architected and shipped a full-stack platform (React, Flask, MySQL REST backend) consolidating multi-generational family health history for disease-risk prediction.
- Trained and deployed a Random Forest classifier reaching ∼82% accuracy predicting diabetes, cancer, and heart-disease risk from patient and family records.
- Resolved a Lucide-React dependency conflict and cross-browser sidebar-icon rendering failures, stabilizing the release for a 20-user pilot group.

GenMouse — Gesture-Controlled Virtual Mouse **Jun 2025**
Python · Keras · OpenCV · MediaPipe

- Built a real-time computer-vision mouse controller using a custom CNN trained on hand-landmark data, achieving >90% gesture-classification accuracy.
- Mapped 6+ gestures to OS-level mouse events (click, scroll, drag) via PyAutoGUI, sustaining ∼50 ms end-to-end response latency.

EDUCATION

Graphic Era Hill University, Bhimtal **Aug 2023 – Jun 2027**
B.Tech, Computer Science Engineering *SGPA: 8.38 / 10*

Hermann Gmeiner School, Bhimtal **2023 / 2021**
Intermediate (92.6%) · Matriculation (94.8%)

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI & Stanford Online (Andrew Ng), Coursera (2026)
- AWS Cloud Practitioner Essentials — AWS Training & Certification (2026)
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — Oracle University

LEADERSHIP & ACHIEVEMENTS

- Led a 4-person team as Team Leader in the Smart India Internal Hackathon 2025, coordinating scope and delivery under a fixed deadline.
- Solved 200+ Data Structures & Algorithms problems across LeetCode and GeeksforGeeks.
- NCC 'A' Certificate (Grade A) recipient.
- Awarded the Reliance Undergraduate Scholarship and Vivo Scholarship (2023) for academic merit and technology-driven project contributions.